

```

%% Efficient Sample Rate Converter using Polyphase Structure
% Polyphase Sample Rate Converter
% L = 3, M = 2
% FIR Length = 36 taps

```

```

clc;
clear;
close all;

```

```

%% Parameters

```

```

L = 3;           % Upsampling factor
M = 2;           % Downsampling factor
N = 36;          % FIR filter length

```

```

Fs = 1000;        % Original Sampling Frequency
t = 0:1/Fs:1;

```

```

%% Input Signal

```

```

x = sin(2*pi*50*t) + 0.5*sin(2*pi*150*t);

```

```

fprintf('Original Sampling Frequency = %.2f Hz\n',Fs);

```

Original Sampling Frequency = 1000.00 Hz

```

%% Design FIR Low-pass Filter

```

```

cutoff = 1/max(L,M);

```

```

h = fir1(N-1,cutoff);

```

```

%% -----

```

```

%% DIRECT IMPLEMENTATION

```

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%% -----

```

```

% Step 1: Upsample

```

```

x_up = upsample(x,L);

```

```

% Step 2: Filter

```

```

y_filter = filter(h,1,x_up);

```

```

% Step 3: Downsample

```

```

y_direct = downsample(y_filter,M);

```

```

Fs_out = Fs*L/M;

```

```

fprintf('Output Sampling Frequency = %.2f Hz\n',Fs_out);

```

Output Sampling Frequency = 1500.00 Hz

```

%% -----

```

```

%% POLYPHASE IMPLEMENTATION
%% -----

% Polyphase decomposition
poly = cell(L,1);

for k = 1:L
    poly{k} = h(k:L:end);
end

% Polyphase filtering
Nout = floor((length(x)*L)/M);

y_poly = zeros(1,Nout);

index = 1;

for n = 1:Nout

    phase = mod((n-1)*M,L) + 1;

    inputIndex = floor(((n-1)*M)/L)+1;

    coeff = poly{phase};

    temp = 0;

    for k = 1:length(coeff)

        if inputIndex-k+1 > 0
            temp = temp + coeff(k)*x(inputIndex-k+1);
        end

    end

    y_poly(index)=temp;
    index=index+1;

end

%% -----
%% Plot Signals
%% -----

figure;

subplot(3,1,1)
plot(x)
title('Original Signal')
xlabel('Samples')

```

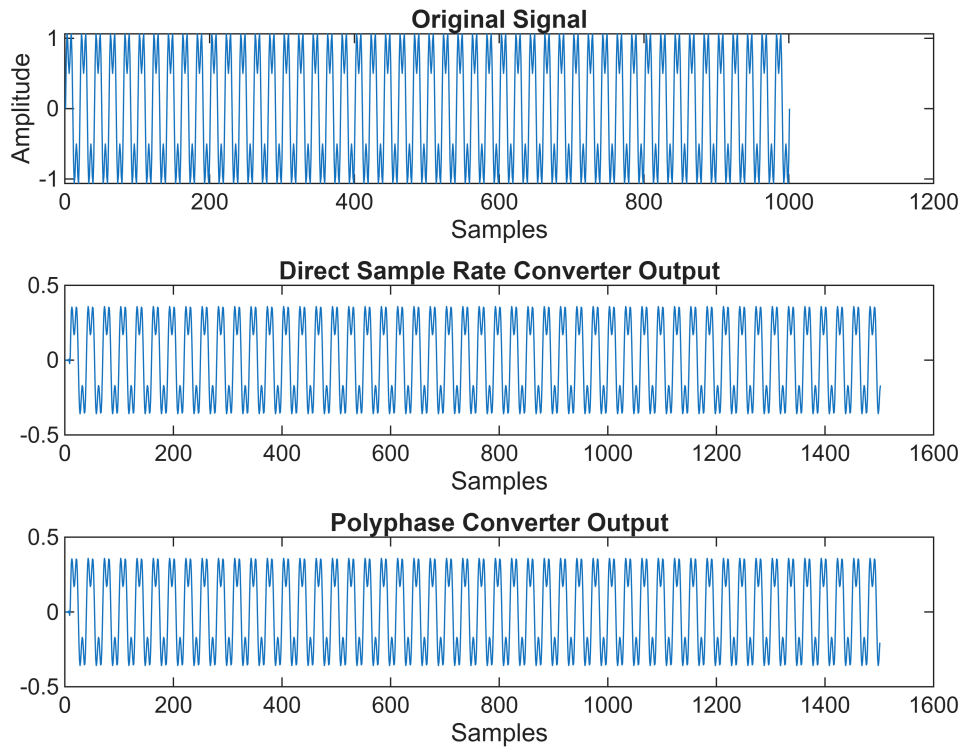
```

ylabel('Amplitude')

subplot(3,1,2)
plot(y_direct)
title('Direct Sample Rate Converter Output')
xlabel('Samples')

subplot(3,1,3)
plot(y_poly)
title('Polyphase Converter Output')
xlabel('Samples')

```

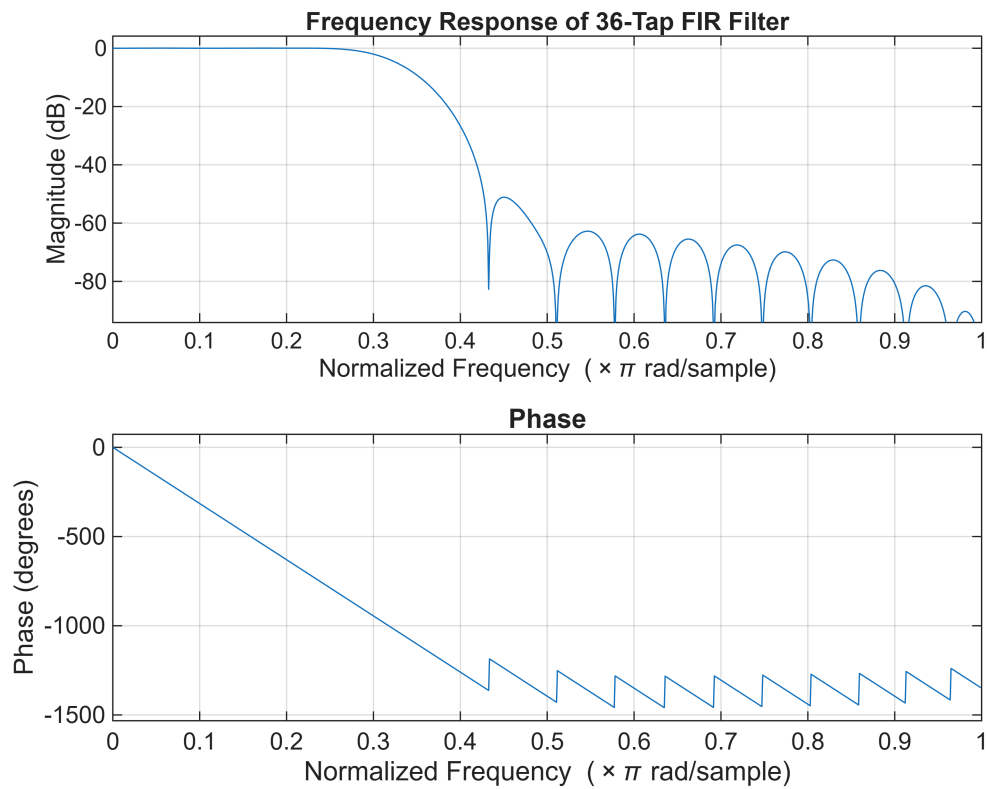


```

%% -----
%% Frequency Response of FIR Filter
%% -----

figure
freqz(h,1,1024)
title('Frequency Response of 36-Tap FIR Filter')

```



```
%% -----
%% Compare Outputs
%% -----

minLen = min(length(y_direct),length(y_poly));

error = y_direct(1:minLen)-y_poly(1:minLen);

fprintf('\nMaximum Difference Between Outputs = %e\n',max(abs(error)));
```

Maximum Difference Between Outputs = 1.665335e-16

```
%% -----
%% Computational Complexity
%% -----

% Direct Implementation
direct_mult = length(h)*length(x_up);

% Polyphase Implementation
poly_mult = (length(h)/L)*Nout;

fprintf('\n----- Computational Complexity -----\n');
```

----- Computational Complexity -----

```
fprintf('Direct FIR Multiplications = %.0f\n',direct_mult);
```

Direct FIR Multiplications = 108108

```
fprintf('Polyphase FIR Multiplications = %.0f\n',poly_mult);
```

Polyphase FIR Multiplications = 18012

```
saving = ((direct_mult-poly_mult)/direct_mult)*100;
```

```
fprintf('Reduction = %.2f %%\n',saving);
```

Reduction = 83.34 %

```
%% -----  
%% Frequency Spectrum Comparison  
%% -----
```

figure

```
subplot(2,1,1)
```

```
NFFT = 2048;
```

```
f1 = linspace(-Fs_out/2,Fs_out/2,NFFT);
```

```
plot(f1,fftshift(abs(fft(y_direct,NFFT))))
```

```
title('Spectrum of Direct Converter')
```

```
xlabel('Frequency (Hz)')
```

```
ylabel('|Y(f)|')
```

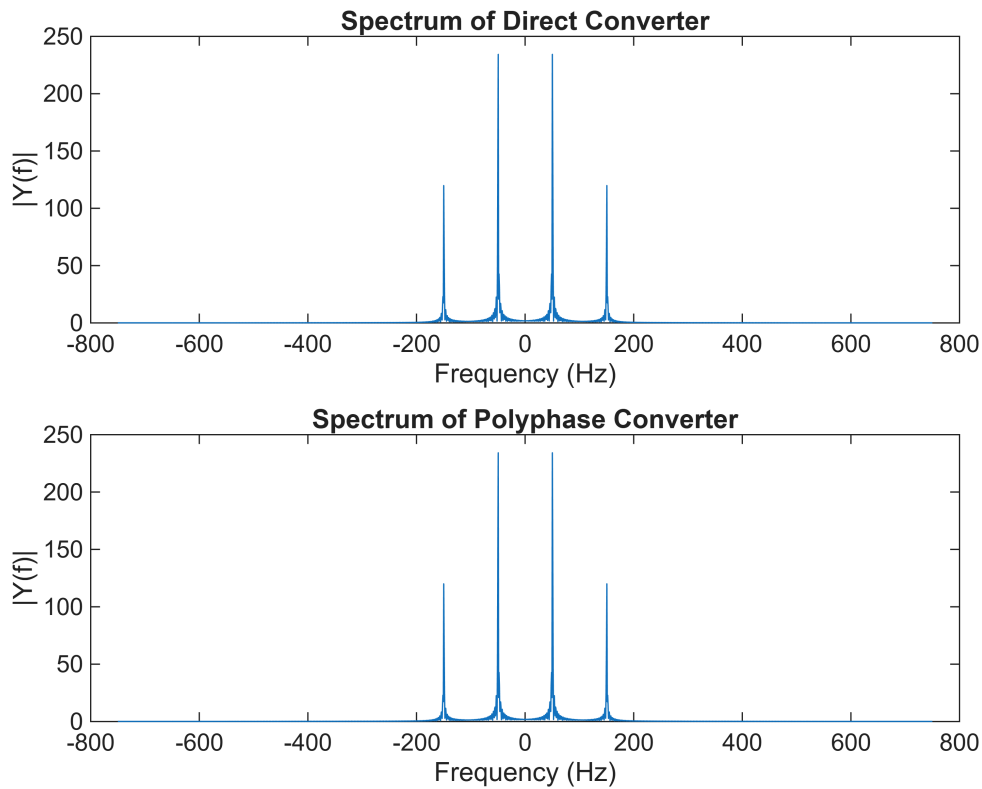
```
subplot(2,1,2)
```

```
plot(f1,fftshift(abs(fft(y_poly,NFFT))))
```

```
title('Spectrum of Polyphase Converter')
```

```
xlabel('Frequency (Hz)')
```

```
ylabel('|Y(f)|')
```



```
%% -----
%% Sampling Rate Verification
%% -----
```

```
fprintf('\nSampling Rate Verification\n');
```

Sampling Rate Verification

```
fprintf('Input Sampling Rate = %.2f Hz\n',Fs);
```

Input Sampling Rate = 1000.00 Hz

```
fprintf('Expected Output Rate = Fs*(L/M)= %.2f Hz\n',Fs*L/M);
```

Expected Output Rate = Fs*(L/M)= 1500.00 Hz

```
fprintf('Verified Output Rate = %.2f Hz\n',Fs_out);
```

Verified Output Rate = 1500.00 Hz

```
disp('Sample Rate Conversion Completed Successfully.')
```

Sample Rate Conversion Completed Successfully.

